

7.8 Mechanical Engineering

FreeStudy

www.freestudy.co.uk/home.htm

This site is a free online tutorial site primarily related to Mechanical Engineering mapped to the Engineering Council's (UK) exams that enables you to achieve Chartered Engineer Status. The contents of the tutorials are pretty exhaustive, and you will need to choose (or skip) the areas of study depending on what is of particular relevance to you. You can view the courses by different classes or by course modules.

Strength of Materials

<http://physics.uwstout.edu/StatStr/index.htm>

There are two introductory courses on this site—Strength of Materials and Statics and Strength of Materials. The emphasis of the course is more on the practical side and less on the theoretical, so it's a good resource for those looking to get a greater insight into the practical applications of various theoretical concepts. A whole range of topics are covered, from Statics of Simple Structures, Columns, Combined Stresses, and Mohr's Circle. There are class lectures, examples, problem assignments, sample tests and their solutions and much more. Rather than treat this site as your sole resource it is advisable to use it as a companion to your textbook material.

Fluid Mechanics

Online Course for Civil Engineers

www.efm.leeds.ac.uk/CIVE/CIVE1400/course.html

This site contains a well-structured online course with a primary focus on civil engineering students. There are detailed course notes, problems with their solutions along with previous exam papers (University of Leeds).